# POLS 6381: Homework Template

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## 0. Loading Packages

```
rm(list=ls()) # Start clean: remove all objects from the workspace

## Libraries
library(tinytex); library(ggplot2); library(ggpubr); library(stargazer)
library(haven);   library(faraway); library(MASS);   library(aod)
library(lmtest);  library(dplyr);   library(AER);    library(glmx)
library(heatmapFit); library(maxLik); library(reshape2); library(foreign);library(car);

options(scipen=100); options(digits = 4)

## Working directory
setwd("/Users/songeunlee/Documents/GitHub/rmarkdown-homework-template")
```

## Load the Data

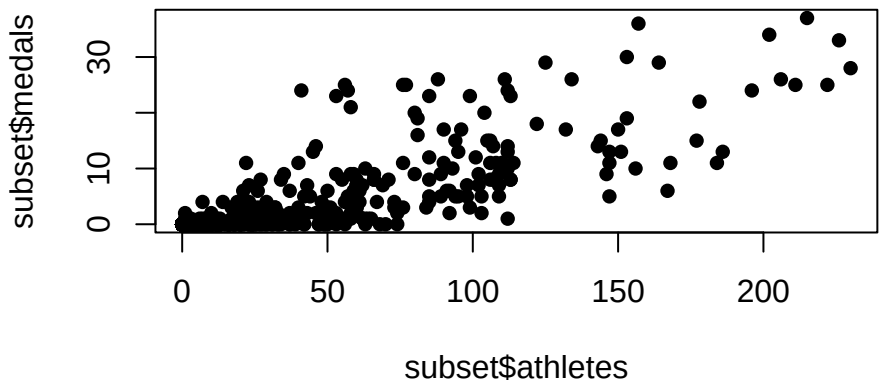
```
# Make sure the data file is saved in the same folder as this .Rmd file
data <- read_dta("Olympics_HW.dta")

# Inspect variable names
names(data)
```

```
 [1] "ID"            "year"          "host"          "country"
 [5] "temp"          "precipitation" "elevation"     "gold"
 [9] "silver"        "bronze"        "population"    "GDP"
[13] "participate"   "medals"        "athletes"      "time"
```

```
# Keep observations with non-missing GDP
subset <- data[complete.cases(data$GDP), ]
```

## 1. Produce a scatterplot with athletes on the horizontal axis and medals on the vertical axis.(3pts)

```
plot(subset$athletes, subset$medals, pch=16) # plot(log(athletes+1), log(medals+1), pch=16)
```




**2. Estimate the following bivariate regression model using ordinary least squares:**

$$medals_i = \beta_0 + \beta_1 athletes_i + \epsilon_i$$

**(a) Examine the coefficients; how does increasing the number of athletes by 1 impact the number of medals a country wins?**

```r
# Estimate the bivariate OLS regression
model1 <- lm(medals ~ athletes, data = subset)

# Display regression results
summary(model1)
```

**Answer: The estimated coefficient for athletes is 0.1145, which means that increasing the number of athletes by one is associated with an increase of about 0.115 medals, on average.**

```
Call:
lm(formula = medals ~ athletes, data = subset)

Residuals:
    Min      1Q  Median      3Q     Max 
-12.751  -0.315   0.258   0.373  19.677 

Coefficients:
            Estimate Std. Error t value             Pr(>|t|)    
(Intercept) -0.37253    0.10845   -3.43              0.00062 ***
athletes     0.11451    0.00251   45.60 < 0.0000000000000002 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.99 on 964 degrees of freedom
Multiple R-squared:  0.683, Adjusted R-squared:  0.683 
F-statistic: 2.08e+03 on 1 and 964 DF,  p-value: <0.0000000000000002
```

```r
# Extract slope coefficient
coef(model1)["athletes"]
```

```
athletes 
  0.1145 
```

**(b) Examine the t statistic and the p value; is the slope coefficient statistically significant ($p < .05$)?**

```r
round(summary(model1)$coefficients, 4)
```

**Answer: Yes, the slope coefficient is statistically significant. The t value is 45.60 and the p-value is less than 0.001, which is far below the 0.05 significance level. Therefore, we reject the null hypothesis that the slope coefficient equals zero.**

```
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  -0.3725     0.1085  -3.435   0.0006
athletes      0.1145     0.0025  45.601   0.0000
```

**(c) Examine the model fit statistics; how much of the variation in medals can be accounted for by differences in the number of athletes?**

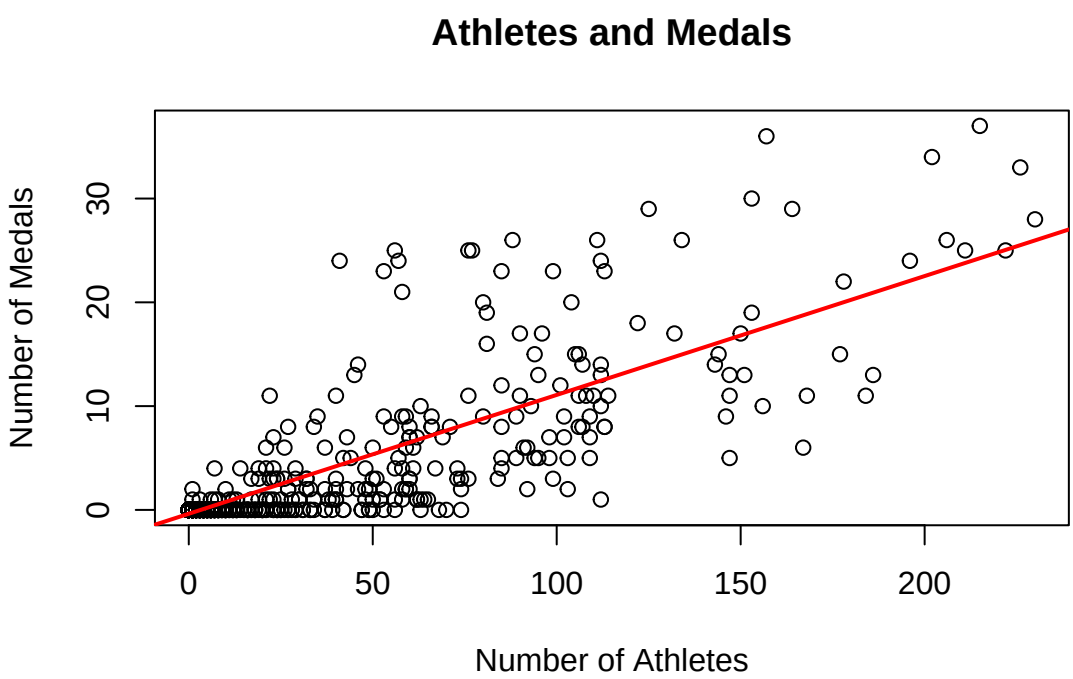
```r
summary(model1)$r.squared
```

**Answer: The R-squared value is 0.6833, which means that about 68.3% of the variation in medals can be explained by differences in the number of athletes.**

```
[1] 0.6833
```

**(d) Add a regression line from this bivariate model to the scatterplot you produced earlier. (3pts)**

```r
plot(subset$athletes, subset$medals,
     xlab = "Number of Athletes",
     ylab = "Number of Medals",
     main = "Athletes and Medals")

abline(model1, col = "red", lwd = 2)
```




**Answer**

### 3. What is the correlation (r) between athletes and GDP? What is the R2 from a supplemental regression in which athletes is the dependent variable and GDP is the independent variable? (3pts)

```r
cor(subset$GDP, subset$athletes)
```

```
[1] 0.3385
```

```r
question3 <- lm(subset$athletes ~ subset$GDP)
summary(question3)
```

```

Call:
lm(formula = subset$athletes ~ subset$GDP)

Residuals:
    Min      1Q  Median      3Q     Max 
-112.87  -13.52  -11.67   -0.38  209.58 

Coefficients:
            Estimate Std. Error t value             Pr(>|t|)    
(Intercept)   11.425      1.385    8.25  0.00000000000000052 ***
subset$GDP     7.192      0.644   11.17 < 0.0000000000000002 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 36.1 on 964 degrees of freedom
Multiple R-squared:  0.115, Adjusted R-squared:  0.114 
F-statistic:  125 on 1 and 964 DF,  p-value: <0.0000000000000002
```